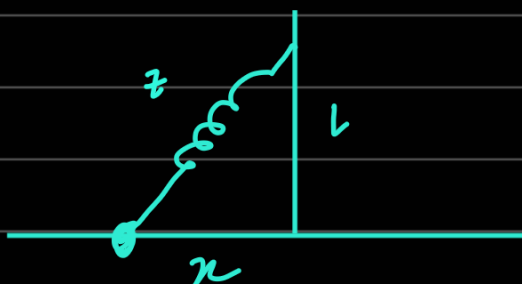


1. Frequency of oscillations I: Find the frequency of oscillations of a particle of mass m which is attached to a spring whose other end is fixed at a distance l and that is

- (a) Free to move along a line,
- (b) Moving on a circle of radius r .

Assume that a force F is required to extend the spring to the length l .

(i)
a)



$$F = kL \Rightarrow k = \frac{F}{L}$$

$$\delta L = \sqrt{l^2 + x^2} - l$$

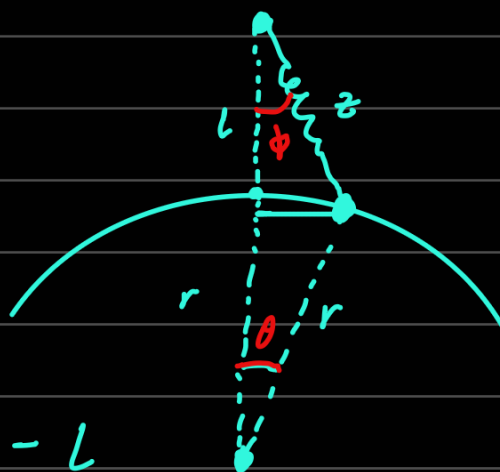
$$\delta L = \frac{x^2}{2l}$$

$$U(x) = \frac{1}{2} k x^2 = \frac{1}{2} k (l + \delta L)^2 = \frac{k l^2}{2} + k l \delta L + \frac{k}{2} \delta L^2$$

$$= \frac{k x^2}{2} \Rightarrow \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{F}{m l}}$$

b) $F = kL$

$$z^2 = (l+r)^2 + r^2 - 2r(l+r)\cos\theta$$



$$\delta L = z - l$$

$$= l \sqrt{\left(\frac{l+r}{l}\right)^2 + \frac{r^2}{l^2} - \frac{2r}{l} \left(\frac{l+r}{l}\right) \cos\theta} - l$$

$$= l \sqrt{1 + x^2 + 2x + x^2 - 2x(1+x)\left(1 - \frac{\theta^2}{2}\right)} - l$$

$$= l \sqrt{1 + x(1+x)\theta^2} - l$$

$$\delta L = \frac{l x (1+x)}{2} \theta^2$$

$$U = k l \delta L = F \delta L = \frac{F l x (1+x)}{2} \theta^2 = \frac{F l x (1+x)}{2 r^2} s^2 = \frac{1}{2} k s^2$$

$$\Rightarrow \omega = \sqrt{\frac{F l x (1+x)}{m r^2}} = \sqrt{\frac{F r (r+l)}{m l r}}$$

4. Forced oscillations: Assuming that, at time $t = 0$, a system is at rest in equilibrium (i.e. $x(0) = \dot{x}(0) = 0$), determine the forced oscillations of the system under a force $F(t)$ of the following forms: (a) $F = F_0$, (b) $F = at$, (c) $F = F_0 \exp -(\alpha t)$, and (d) $F = F_0 \exp -(\alpha t) \cos(\beta t)$, with F_0 , a , α and β being constants.

$$\xi = e^{i\omega t} \int_0^t \frac{F(t')}{m} e^{-i\omega t'} dt'$$

($\xi_0 = 0$ because $x = \dot{x} = 0$ at $t=0$)

a) $\xi = e^{i\omega t} \int_0^t \frac{F_0}{m} e^{-i\omega t'} dt'$

$$\xi = \frac{1}{m} e^{i\omega t} \frac{F_0 e^{-i\omega t'}}{-i\omega} \Big|_0^t = -\frac{iF_0}{\omega} e^{i\omega t} (1 - e^{-i\omega t})$$

$$\xi = \frac{-iF_0}{m\omega} (e^{i\omega t} - 1) = \frac{-iF_0}{\omega} (\cos \omega t - 1 + i \sin \omega t) = \dot{x} + i\omega x$$

$$\Rightarrow \boxed{x = \frac{F_0}{m\omega^2} (1 - \cos \omega t)}$$

b) $\xi = \frac{ae^{i\omega t}}{m} \int_0^t t' e^{-i\omega t'} dt' = \frac{ae^{i\omega t}}{m} \left(\frac{it' e^{-i\omega t'}}{\omega} + \frac{e^{-i\omega t'}}{\omega^2} \right) \Big|_0^t$

$$\xi = \frac{ae^{i\omega t}}{m} \left(\frac{it e^{-i\omega t}}{\omega} + \frac{e^{-i\omega t} - 1}{\omega^2} \right)$$

$$\xi = \frac{a}{m} \left(\frac{it}{\omega} + \frac{1 - e^{i\omega t}}{\omega^2} \right)$$

$$\xi = \frac{a}{m} \left(\frac{1 - \cos \omega t}{\omega^2} \right) + ia \left(\frac{t}{\omega} - \frac{\sin \omega t}{\omega^2} \right) = \dot{x} + i\omega x$$

$$\Rightarrow \boxed{x = \frac{a}{m\omega^3} (\omega t - \sin \omega t)}$$

c) $\xi = \frac{e^{i\omega t}}{m} \int_0^t F_0 e^{-(i\omega + \alpha)t'} dt' = \frac{F_0 e^{i\omega t}}{m(\alpha + i\omega)} (1 - e^{-(i\omega + \alpha)t})$

$$= \frac{F_0(\alpha - i\omega)}{\alpha^2 + \omega^2} (e^{i\omega t} - e^{-\alpha t})$$

$$\xi = \frac{F_0}{a^2 + \omega^2} (a - i\omega) (\cos \omega t - e^{-at} + i \sin \omega t)$$

$$\xi = \frac{F_0}{a^2 + \omega^2} \left(\dots + i (a \sin \omega t - \omega \cos \omega t + \omega e^{-at}) \right) = \ddot{x} + i\omega \dot{x}$$

$$\Rightarrow \boxed{x = \frac{F_0}{m(a^2 + \omega^2)} \left(\frac{a}{\omega} \sin \omega t - \cos \omega t + e^{-at} \right)}$$

$$d) \quad \xi = e^{i\omega t} \int_0^t F_0 e^{-at'} \left(\frac{e^{i\beta t'} + e^{-i\beta t'}}{2} \right) e^{-i\omega t'} dt'$$

$$\xi = \frac{F_0 e^{i\omega t}}{2} \int_0^t \left(e^{(i\beta - i\omega - a)t'} + e^{(-i\beta - i\omega - a)t'} \right) dt'$$

$$\xi = \frac{F_0 e^{i\omega t}}{2} \left(\frac{e^{(i\beta - i\omega - a)t'}}{i\beta - i\omega - a} \Big|_0^t + \frac{e^{(-i\beta - i\omega - a)t'}}{-i\beta - i\omega - a} \Big|_0^t \right)$$

$$\xi = \frac{F_0 e^{i\omega t}}{2} \left(\frac{1 - e^{(i\beta - i\omega - a)t}}{a + i(\omega - \beta)} + \frac{1 - e^{(-i\beta - i\omega - a)t}}{a + i(\omega + \beta)} \right)$$

$$\xi = \frac{F_0 e^{i\omega t}}{2} \left(\frac{(1 - e^{(i\beta - i\omega - a)t})(a - i(\omega - \beta))}{a^2 + (\omega - \beta)^2} + \frac{(1 - e^{(-i\beta - i\omega - a)t})(a - i(\omega + \beta))}{a^2 + (\omega + \beta)^2} \right)$$

$$\xi = \frac{F_0 e^{i\omega t}}{2} \left(\frac{a - i(\omega - \beta) - a e^{(i\beta - i\omega - a)t} + i(\omega - \beta) e^{(i\beta - i\omega - a)t}}{a^2 + (\omega - \beta)^2} + \frac{a - i(\omega + \beta) - a e^{(-i\beta - i\omega - a)t} + i(\omega + \beta) e^{(-i\beta - i\omega - a)t}}{a^2 + (\omega + \beta)^2} \right)$$

$$\xi = \frac{F_0}{2} \left(\frac{a e^{i\omega t} - i(\omega - \beta) e^{i\omega t} - a e^{i(\beta - \omega)t} + i(\omega - \beta) e^{i(\beta - \omega)t}}{a^2 + (\omega - \beta)^2} \right.$$

$$\left. + \frac{a e^{i\omega t} - i(\omega + \beta) e^{i\omega t} - a e^{(-i\beta - \omega)t} + i(\omega + \beta) e^{(-i\beta - \omega)t}}{a^2 + (\omega + \beta)^2} \right)$$

$$\xi = \frac{F_0}{2} \left(\frac{a \cos \omega t + i a \sin \omega t - (\omega - \beta) (i \cos \omega t - \sin \omega t) - a e^{-\alpha t} (\cos \beta t + i \sin \beta t)}{a^2 + (\omega - \beta)^2} + \frac{a \cos \omega t + i a \sin \omega t - (\omega + \beta) (i \cos \omega t - \sin \omega t) - a e^{-\alpha t} (\cos \beta t - i \sin \beta t)}{a^2 + (\omega + \beta)^2} \right)$$

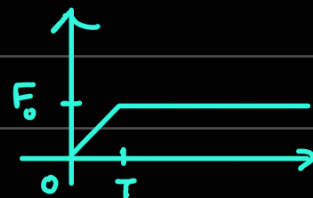
⇒

$$x = \frac{F_0}{2m\omega} \left(\frac{a \sin \omega t - (\omega - \beta) \cos \omega t - a e^{-\alpha t} \sin \beta t + (\omega - \beta) e^{-\alpha t} \cos \beta t}{a^2 + (\omega - \beta)^2} + \frac{a \sin \omega t - (\omega + \beta) \cos \omega t + a e^{-\alpha t} \sin \beta t + (\omega + \beta) e^{-\alpha t} \cos \beta t}{a^2 + (\omega + \beta)^2} \right)$$

5. Final amplitude: Assuming that a system is at rest in equilibrium up to time $t = 0$, determine the final amplitude for the oscillations of the system under: (a) a force which is zero for $t < 0$, $F_0 t/T$ for $0 < t < T$, and F_0 for $t > T$, (b) a constant force F_0 that acts for a finite time T , and (c) forces (i) $F_0 t/T$ and (ii) $F_0 \sin(\omega t)$, which act between $t = 0$ and $t = T$.

a) For $(0, T)$: $\xi = \frac{e^{i\omega t}}{m} \int_0^T \frac{F_0}{T} t' e^{-i\omega t'} dt' + \cancel{\xi_0} \rightarrow 0$

From 'b' above, with $a = \frac{F_0}{T}$,



$$x = \frac{F_0}{Tm\omega^3} (\omega t - \sin \omega t) \quad t \in (0, T)$$

For $t > T$, $x = A \sin \omega(t-T) + B \cos \omega(t-T) + \frac{F_0}{m\omega^2}$

→ shifting due to constant F

For continuity, $x(T^-) = x(T^+)$ and $\dot{x}(T^-) = \dot{x}(T^+)$

$$\Rightarrow \frac{F_0}{Tm\omega^3} (\omega T - \sin \omega T) = B + \frac{F_0}{m\omega^2}$$

x

$$\text{and } \frac{F_0}{Tm\omega^3} (\omega - \omega \cos \omega T) = A\omega$$

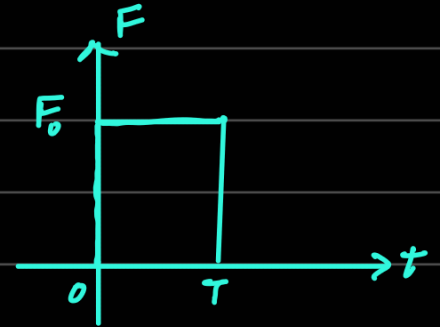
$\dot{x}$

$$\Rightarrow A = \frac{F_0}{Tm\omega^3} (1 - \cos \omega T) \quad \text{and} \quad B = -\frac{F_0 \sin \omega T}{Tm\omega^3}$$

$$\therefore x(t) = \begin{cases} \frac{F_0}{Tm\omega^3} (\omega t - \sin \omega t) & t \in [0, T] \\ \frac{F_0}{Tm\omega^3} (1 - \cos \omega T) \sin \omega(t-T) - \frac{F_0 \sin \omega T}{Tm\omega^3 + \frac{F_0}{m\omega^2}} \cos \omega(t-T) & t \geq T \end{cases}$$

b) For $t \in [0, T]$,

$$x = \frac{F_0}{m\omega^2} (1 - \cos \omega t), \quad \dot{x} = \frac{F_0 \sin \omega t}{m\omega}$$



For $t \geq T$,

$$x = A \sin \omega(t-T) + B \cos \omega(t-T)$$

$$\dot{x} = A\omega \cos \omega(t-T) - B\omega \sin \omega(t-T)$$

For continuity at $t=T$,

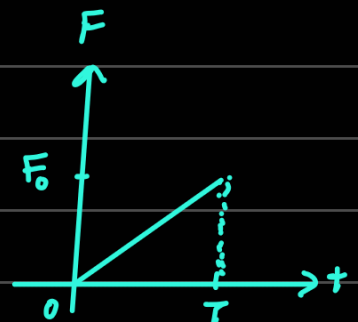
$$B = \frac{F_0}{m\omega^2} (1 - \cos \omega T)$$

$$A\omega = \frac{F_0 \sin \omega T}{m\omega} \Rightarrow A = \frac{F_0 \sin \omega T}{m\omega^2}$$

$$\therefore x(t) = \begin{cases} \frac{F_0}{m\omega^2} (1 - \cos \omega t) & t \in [0, T] \\ \frac{F_0 \sin \omega T}{m\omega^2} \sin \omega(t-T) + \frac{F_0 (1 - \cos \omega T)}{m\omega^2} \cos \omega(t-T) & t \geq T \end{cases}$$

c) i) $x = \frac{F_0}{Tm\omega^3} (\omega t - \sin \omega t) \quad t \in [0, T]$

$$\dot{x} = \frac{F_0}{Tm\omega^2} (1 - \cos \omega t)$$



For $t \geq T$,

$$x = A \sin \omega(t-T) + B \cos \omega(t-T)$$

$$\dot{x} = A\omega \cos \omega(t-T) - B\omega \sin \omega(t-T)$$

For continuity, $\frac{F_0}{Tm\omega^3} (\omega T - \sin \omega T) = B$

$$\frac{F_0}{Tm\omega^3} (1 - \cos \omega T) = A$$

$$\therefore x(t) = \begin{cases} \frac{F_0}{Tm\omega^3} (\omega t - \sin \omega t) & t \in [0, T] \\ \frac{F_0}{Tm\omega^3} (1 - \cos \omega T) \sin \omega(t-T) + \frac{F_0}{Tm\omega^3} (\omega T - \sin \omega T) \cos \omega(t-T) & t \geq T \end{cases}$$

ii) For $t \in [0, T]$,

$$\xi_t = e^{i\omega t} \int_0^t \frac{F_0 (e^{i\omega t'} - e^{-i\omega t'})}{2mi} \cdot e^{-i\omega t'} dt'$$

$$\xi_t = \frac{F_0}{2mi} \int_0^t (1 - e^{-2i\omega t'}) dt' = \frac{F_0}{2mi} \left(t' + \frac{e^{-2i\omega t'}}{2i\omega} \right) \Big|_0^t$$

$$= \frac{F_0}{2mi} \left(t + \frac{e^{-2i\omega t} - 1}{2i\omega} \right)$$

$$= -\frac{F_0}{2m} \left(it + \frac{e^{-2i\omega t} - 1}{2\omega} \right)$$

$$\xi_t = -\frac{F_0}{2m} \left(it + \frac{\cos 2\omega t - i \sin 2\omega t - 1}{2\omega} \right)$$

$$\Rightarrow x = \frac{F_0}{2m\omega} \left(\frac{\sin 2\omega t}{2\omega} - t \right), \quad \dot{x} = \frac{F_0}{2m\omega} (\cos 2\omega t - 1)$$

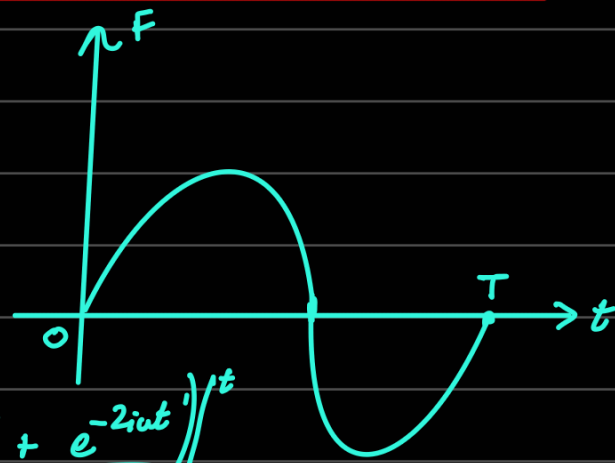
For $t \geq T$,

$$x = A \sin \omega(t-T) + B \cos \omega(t-T)$$

$$\dot{x} = A \omega \cos \omega(t-T) - B \omega \sin \omega(t-T)$$

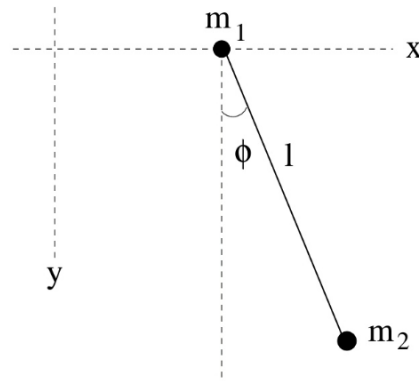
For continuity
at $t = T = \frac{2\pi}{\omega}$,

$$-\frac{F_0 \pi}{m\omega^2} = B, \quad A = 0$$



$$\therefore x(t) = \begin{cases} \frac{F_0}{2m\omega} \left(\frac{\sin 2\omega t}{2\omega} - t \right) & t \in [0, \tau] \\ -\frac{F_0 \pi}{m\omega^2} \cos \omega(t-\tau) & t \geq \tau \end{cases}$$

2. Frequency of oscillations II: Find the frequency of small oscillations of a pendulum whose point of support carries a mass m_1 and is free to move horizontally as shown in the figure on the left below.



$$m_1: (x, 0) \quad m_2: \left(x + l\phi, l - \frac{l\phi^2}{2} \right) \quad (\dot{x} + l\dot{\phi}, -l\phi\dot{\phi})$$

$$\mathcal{L} = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 \left(\dot{x}^2 + 2l\dot{x}\dot{\phi} + l^2\dot{\phi}^2 + l^2\phi^2\dot{\phi}^2 \right) - m_2 g l \frac{\phi^2}{2}$$

$$\mathcal{L} = \frac{1}{2} (m_1 + m_2) \dot{x}^2 + m_2 l \dot{x}\dot{\phi} + \frac{1}{2} m_2 l^2 \dot{\phi}^2 (1 + \phi^2) + m_2 g l \frac{\phi^2}{2}$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = m_2 l^2 \dot{\phi}^2 \phi - m_2 g l \phi \quad \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = m_2 l \dot{x} + m_2 l^2 \dot{\phi} (1 + \phi^2)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = (m_1 + m_2) \dot{x} + m_2 l \dot{\phi}$$

$$m_2 l \ddot{x} + m_2 l^2 \ddot{\phi} (1 + \phi^2) = -m_2 l^2 \dot{\phi}^2 \phi - m_2 g l \phi$$

$$(m_1 + m_2) \ddot{x} + m_2 l \ddot{\phi} = 0$$

$$\ddot{x} = -\frac{m_2 l}{m_1 + m_2} \ddot{\phi}$$

$$\frac{-m_2^2 l^2}{m_1 + m_2} \ddot{\phi} + m_2 l^2 \ddot{\phi} (1 + \phi^2) + m_2 l^2 \dot{\phi}^2 \phi = -m_2 l g \phi$$

$$\frac{-m_2}{m_1 + m_2} \ddot{\phi} + \ddot{\phi} + \ddot{\phi} \phi^2 + \dot{\phi}^2 \phi = -\frac{g \phi}{l}$$

$$\frac{m_1}{m_1 + m_2} \ddot{\phi} = -\frac{g}{l} \phi$$

$$\therefore \omega = \sqrt{\frac{g}{l} \left(\frac{m_1 + m_2}{m_1} \right)}$$